Jane Street TDOE

Two days. One focused plan.

First live interview: probability, counting, ratios and numerical problem solving.

24	02	24
timed questions	fresh mock interviews	worked solutions

START HERE

Read the plan on page 3. Complete the 14 Core drills, then take the two mocks without opening the hints or solutions. Use the four Stretch drills only after you have repaired your weak areas.

Navigate Evidence 2 | Plan 3 | Toolkit 4
Drills 5-10 | Mocks 11-12 | Hints 13-14
Solutions 15-22 | Final review 23 | Sources 24-25

Based on all 33 reports read in your Glassdoor tab, official Jane Street material and additional role-specific candidate accounts. Reported questions, reconstructed prompts and original practice are labelled separately.

01 / RESEARCH FINDINGS

Prepare for the right round

The strongest evidence is role-specific. Broad quant-trader puzzle lists are a lower priority.

Jane Street describes a take-home exercise, an initial Zoom focused on problem solving and critical thinking, a more technical Zoom, and a final round. It encourages candidates to test their answers and clarify unfamiliar information. [O1]

The 33 TDOE reports already read in this chat repeatedly place maths and logic before coding. Additional TDOE intern and Wall Street Oasis accounts reinforce that pattern. Some candidates count the take-home as round one. This pack targets the **first live problem-solving interview**. [G0] [W1] [W2]

HIGHEST PRIORITY

Conditional probability, counting and palindromes, ratios and invariants. Repeated concrete examples support these families.

NEXT PRIORITY

Expected values, waiting times, speed-distance-time and digit constraints. Practise changing assumptions and follow-ups.

KEEP IN RESERVE

Simple state-based recursion and age or missing-digit puzzles. These are useful extensions, not a verified round-one syllabus.

LATER-STAGE MATERIAL

Top-k trades, sorting and systems/order-book design appear mainly in later interviews. Spend these two days on the first-live-round task unless your invitation says otherwise.

HOW TO READ THE LABELS

Reported: a recognisable candidate-reported question, paraphrased.
Reported family: a reported structure with assumptions or follow-ups added.
Reconstructed: a complete practice prompt built from an incomplete report.
Official-guide practice: a setup from Jane Street's public trading guide.
Authored drill / mock: practice created for this pack, not a claim that Jane Street asked it.

Evidence limits: anonymous recollections are unverified, some wording is duplicated, posting dates may differ from interview dates, and adjacent TDO/TDOE/intern titles can have different processes. Priority labels are editorial choices, not estimated probabilities of receiving a question.

02 / YOUR 48-HOUR PLAN

Practise, review, then repeat

About 5 focused hours each day, plus breaks. Do not try to fill every waking hour.

DAY 1 / BUILD RELIABLE METHODS

25 min	Read pages 2-4. Explain each formula in your own words.
70 min	Q01-Q03: invariants, cube cases and free-position counting. Solve aloud, then use hints and solutions.
75 min	Q04-Q06 and Q18: sample spaces, combinations and conditioning. Draw small outcome tables.
55 min	Q10-Q11: expected values, reroll decisions and restart equations.
55 min	Q13 and Q15. Re-solve the three hardest earlier questions with changed numbers.
40 min	Watch the official trading mock video for communication style, then write a short error log. Different role; do not copy a persona. [O3]

DAY 2 / ADAPT UNDER PRESSURE

25 min	Recall formulas and yesterday's mistakes without looking at the answer pages.
70 min	Q07-Q09: waiting-time assumptions and dice restart rules. These merit slow, careful modelling.
45 min	Mock A on page 11. No hints, notes or solution pages.
40 min	Review Mock A. Rebuild any wrong answer from the model, not from memory.
45 min	Mock B on page 12. Speak every assumption and sanity check.
45 min	Review Mock B. If weak areas remain, repair them; otherwise try Q12, Q14, Q16 or Q17.
30 min	Use page 23. Prepare a 60-second explanation of why TDOE and why Jane Street. Stop heavy practice early enough to rest.

Running short? Prioritise Q01-Q07, Q09-Q10, Q13 and one full mock. Keep a mistake log with: the wrong assumption, the corrected model, and one changed-number retest. Core and mock times are practice budgets, not hiring thresholds.

03 / WORKING TOOLKIT

A method you can explain

Use this sequence on every question: clarify, model, solve, check, extend.

1 / CLARIFY THE EXPERIMENT

What is selected uniformly? Are draws independent? Is there replacement? Are leading zeros allowed? What happens on every kind of tie? What exactly is visible or observed?

2 / CHOOSE A SIMPLE REPRESENTATION

List a tiny sample space, draw a timeline, group cases, or define one variable. For palindromes, write the forced pattern before counting. For ratios, identify the unchanged quantity.

3 / COMPUTE WITH THE RIGHT TOOL

Counting: C(n,k) = n!/[k!(n-k)!]. Order matters for labelled dice and digit positions.
Conditioning: P(A | B) = P(A and B)/P(B), for P(B)>0.
Expectation: E[X] = sum of x P(X=x). E[X+Y] = E[X]+E[Y], even without independence.
At least one: 1 - P(none). For n independent trials of success chance p: 1-(1-p)^n.
Restart: E = cost of this attempt + P(restart) x E, when a restart returns to the same state.
Relative motion: add speeds when approaching, subtract when chasing. Use one time unit.

4 / SANITY-CHECK

Probabilities lie in [0,1]. Expected payoffs lie between possible extremes. Distances add to the stated gap. Dry mass is conserved. Try a small or extreme case and check your denominator.

5 / HANDLE THE FOLLOW-UP

Say which assumption changed and which part of your previous solution still applies. If stuck, explain a useful partial result and ask a specific clarifying question.

A USEFUL WAY TO THINK ALOUD

"I will assume the selection is uniform. I will count ordered outcomes because the dice are labelled. Here is a smaller case to check my setup. This result has the right bounds. If the tie rule changes, I would change this term."

These are mathematical tools and practice advice. The official guide is useful background, but is aimed at trading/research rather than this specific TDOE round. [O2]

04 / PRACTICE SET 1 OF 6

Invariants and counting

Attempt first. Follow-ups are part of the exercise; hints and solutions are kept separate.

Q01 / REPORTED / CORE / 5 MIN

The dehydrating strawberry

A strawberry weighs 100 g and is 99% water by mass. Water evaporates until it is 98% water. Its non-water material is unchanged. What is its new mass?

Follow-up Generalise: initial mass M, initial water fraction p, final water fraction q.

Hint p.13 | Solution p.15 [G1]

Q02 / REPORTED / CORE / 10 MIN

The painted cube

Paint every outside face of a 3 x 3 x 3 cube made from 27 unit cubes. Mix the unit cubes, choose one uniformly, and roll it with all six orientations equally likely. You can see its five exposed faces. What is the probability of seeing any paint?

Follow-up What if you can see only the top face? Then generalise the five-visible-faces case to an n x n x n cube, n >= 2.

Hint p.13 | Solution p.15 [G2]

Q03 / REPORTED FAMILY / CORE / 6 MIN

Palindromes: count the free positions

Using 26 letters, how many four-letter strings read the same forwards and backwards? Repetition is allowed. How many five-letter palindromes are there?

Follow-up For length five, how many have no equal adjacent letters? What is the general count for length n over an alphabet of size a?

Hint p.13 | Solution p.15 [G3]

04 / PRACTICE SET 2 OF 6

Define the sample space

Attempt first. Follow-ups are part of the exercise; hints and solutions are kept separate.

Q04 / REPORTED FAMILY / CORE / 7 MIN

A random three-digit number

Choose an integer uniformly from 100 to 999. What is the probability that its first digit is odd, its second digit is even, and its last digit differs from both of the first two?

Follow-up What changes if instead you generate three independent digits from 0 to 9, allowing a leading zero?

Hint p.13 | Solution p.16 [G4]

Q05 / REPORTED / CORE / 5 MIN

Exactly three heads

Flip a fair coin five independent times. What is the probability of exactly three heads?

Follow-up Now suppose you know there was at least one tail. What is the conditional probability of exactly three heads? What if the coin has head probability p?

Hint p.13 | Solution p.16 [G5]

Q06 / RECONSTRUCTED / CORE / 7 MIN

Red and blue balls

A bag has three red and two blue balls. Draw two uniformly without replacement. You are told that at least one is red. What is the probability both are red?

Follow-up Instead, you are told that the first ball drawn is red. What is the probability the second is red? Why do the answers differ?

Hint p.13 | Solution p.16 [G6]

04 / PRACTICE SET 3 OF 6

Waiting and restart rules

Attempt first. Follow-ups are part of the exercise; hints and solutions are kept separate.

Q07 / RECONSTRUCTED / CORE / 8 MIN

Wait for the train or walk?

A train departs exactly every 10 minutes. You arrive at a time uniformly distributed within a departure interval, independently of the timetable. The ride takes 5 minutes; walking takes 10. Compare the expected journey times. What is the probability the train is strictly faster?

Follow-up Consider this policy: wait at most 5 minutes, then walk if no train has arrived. What is its expected total journey time?

Hint p.13 | Solution p.17 [G7]

Q08 / RECONSTRUCTED / CORE / 10 MIN

Why irregular buses make you wait

Successive bus departure gaps are independently 5 or 15 minutes, each with probability 1/2. You arrive at an independent random time in the long run, not immediately after a departure. What is your expected wait?

Follow-up The ride takes 5 minutes and walking takes 10. Which has the smaller expected journey time? Why is "half the average gap" wrong?

Hint p.13 | Solution p.17 [G7] [G8]

Q09 / RECONSTRUCTED / CORE / 12 MIN

Three dice and a restart rule

Three players roll independent fair six-sided dice. A unique highest roll wins. If the highest roll is tied, all players reroll, repeating until there is a winner. What is your probability of winning in one round, and eventually?

Follow-up Now your die is fixed at 4 on every round; the other two still roll fairly. Find both probabilities again. State exactly what the restart rule covers.

Hint p.13 | Solution p.17 [G9]

04 / PRACTICE SET 4 OF 6

Expected values and states

Attempt first. Follow-ups are part of the exercise; hints and solutions are kept separate.

Q10 / OFFICIAL-GUIDE PRACTICE / CORE / 7 MIN

One optional reroll

Roll a fair six-sided die. After seeing its value, you may keep it or discard it and roll once more. If you reroll, you must take the second result. Your payoff is the final value. What strategy maximises the expected payoff, and what is that expectation?

Follow-up What is the largest fee you would pay for the reroll option before the first roll? Assume you care only about expected payoff.

Hint p.13 | Solution p.18 [O2]

Q11 / AUTHORED DRILL / CORE / 6 MIN

How long until the first six?

Roll a fair six-sided die repeatedly until the first 6. What is the expected number of rolls, including the successful roll? What is the probability you need more than three rolls?

Follow-up You have already seen ten failures. What is the expected number of additional rolls?

Hint p.13 | Solution p.18 [O2]

Q12 / AUTHORED DRILL / STRETCH / 10 MIN

Two consecutive heads

Flip a fair coin repeatedly until two heads appear consecutively. What is the expected total number of flips? Count all flips, including tails and the final head.

Follow-up Why is treating disjoint pairs of flips as independent attempts inefficient? Can you set up states for three consecutive heads?

Hint p.13 | Solution p.18 [O2]

04 / PRACTICE SET 5 OF 6

Translate words into structure

Attempt first. Follow-ups are part of the exercise; hints and solutions are kept separate.

Q13 / RECONSTRUCTED / CORE / 7 MIN

Two cyclists approaching each other

Two towns are 60 km apart. One cyclist leaves town A at 12 km/h and another leaves town B at 18 km/h, riding towards each other at constant speeds. When and where do they meet?

Follow-up The second cyclist now starts 30 minutes late. When do they meet, measured from the first start, and how far from A?

Hint p.14 | Solution p.19 [G10]

Q14 / RECONSTRUCTED / STRETCH / 7 MIN

Grandmother and two grandchildren

A grandmother is between 60 and 89 years old. Her two-digit age has tens digit a and units digit b. Her two grandchildren are aged a and b years. All three ages sum to 84. Find their ages.

Follow-up Which details make this version solvable? What would you ask if the total age had not been supplied?

Hint p.14 | Solution p.19 [G11]

Q15 / REPORTED FAMILY / CORE / 9 MIN

Consecutive sums

Find the greatest three-digit positive integer that can be written as a sum of two consecutive positive integers, as a sum of three consecutive positive integers, and as a sum of five consecutive positive integers.

Follow-up Exhibit all three representations. Explain why your answer is the greatest possible.

Hint p.14 | Solution p.19 [G12]

04 / PRACTICE SET 6 OF 6

Counting and information

Attempt first. Follow-ups are part of the exercise; hints and solutions are kept separate.

Q16 / AUTHORED DRILL / STRETCH / 7 MIN

Counting a digit without listing numbers

Write every integer from 1 to 999. How many times does the digit 1 appear in total? Count repeated appearances within a number separately.

Follow-up Does the same answer hold for the digit 0? Explain why a leading-zero trick can fail for zero.

Hint p.14 | Solution p.20

Q17 / RECONSTRUCTED / STRETCH / 8 MIN

The missing digit

A six-digit access code is formed from digits 0-9, and leading zeros are allowed. You know five digits and know exactly which position is missing. Every missing digit is equally likely. Each trial tests one full code. How many trials guarantee a successful entry, and what is the expected number if you never repeat a trial?

Follow-up Instead, you only know that deleting one digit from the code produced 12345. How many distinct candidate codes are possible?

Hint p.14 | Solution p.20 [G13]

Q18 / AUTHORED DRILL / CORE / 7 MIN

Dice: condition before you count

Roll two independent fair six-sided dice, labelled A and B. What is the probability their sum is 8? Given that the sum is 8, what is the probability A shows 6?

Follow-up Given only that at least one die shows 6, what is the probability both show 6?

Hint p.14 | Solution p.20

05 / MOCK A

A 45-minute live rehearsal

Read only this page. Speak aloud. If possible, have someone deliver the follow-ups after your first answer.

Use 35 minutes for the three prompts and follow-ups, then 10 minutes to explain checks and reflect. Suggested per-question times below are flexible. Do not open the hints or solutions during the mock.

Q19 / ORIGINAL MOCK / 8 MIN

Changing water content

A batch of fruit weighs 200 g and is 95% water. After drying, it is 90% water. The dry material is unchanged. Find its final mass and the mass of water lost.

Interviewer follow-up What final water percentage would make the batch weigh 125 g?

Q20 / ORIGINAL MOCK / 12 MIN

Constrained palindromes

An alphabet contains five distinct symbols. How many length-five palindromes can you make if repetition is allowed? Of these, how many contain exactly three distinct symbols?

Interviewer follow-up Choose a palindrome uniformly from all length-five palindromes. What is the probability it uses exactly three symbols?

Q21 / ORIGINAL MOCK / 15 MIN

Information about a sample

A bag contains four green and three orange balls. Two are drawn uniformly without replacement. You are told they have the same colour. What is the probability they are both green?

Interviewer follow-up Now repeat with replacement, keeping the same information. Does the answer increase or decrease?

SELF-ASSESSMENT / NOT A HIRING SCORE

For each prompt, award yourself 0-2 on each: clear assumptions, valid model, correct calculation, and explanation/checks. Record whether you solved independently or needed help. Fix the lowest-scoring skill before repeating.

05 / MOCK B

A 45-minute live rehearsal

Read only this page. Speak aloud. If possible, have someone deliver the follow-ups after your first answer.

Use 35 minutes for the three prompts and follow-ups, then 10 minutes to explain checks and reflect. Suggested per-question times below are flexible. Do not open the hints or solutions during the mock.

Q22 / ORIGINAL MOCK / 8 MIN

Another commute

A bus leaves exactly every 8 minutes. You arrive uniformly within a departure interval, independently of the timetable. The bus ride takes 6 minutes and walking takes 11. Which has the smaller expected time, and what is the probability the bus is strictly faster?

Interviewer follow-up If you see the bus leave just as you arrive, should you still use the original expected wait?

Q23 / ORIGINAL MOCK / 15 MIN

The larger die

Roll two independent fair six-sided dice. Your payoff is the larger result. Find its expected value without writing all 36 outcomes individually.

Interviewer follow-up What is the expected payoff if one die is fixed at 4 and the other remains fair?

Q24 / ORIGINAL MOCK / 12 MIN

A delayed cyclist

Two cyclists start from opposite ends of a 50 km road, riding towards each other at 10 and 15 km/h. Find their meeting time and the distance travelled by the slower rider.

Interviewer follow-up The faster rider now starts 20 minutes late. Find the meeting time from the slower rider's start, and the meeting point.

SELF-ASSESSMENT / NOT A HIRING SCORE

For each prompt, award yourself 0-2 on each: clear assumptions, valid model, correct calculation, and explanation/checks. Record whether you solved independently or needed help. Fix the lowest-scoring skill before repeating.

06 / HINTS

One nudge at a time / 1 of 2

Use the smallest hint you need, then return to the question. Do not read ahead for a mock.

Q01

Keep the dry mass fixed. What fraction of the final mass is dry matter?

Q02

Classify cubes by their number of painted faces. When can all paint be hidden?

Q03

Write ABBA and ABCBA. Only count the positions you are free to choose.

Q04

The first digit has nine possible values. The second can be zero. Odd and even digits cannot be equal.

Q05

Choose which three positions contain heads. Then restrict the sample space for the extra information.

Q06

Count unordered pairs for "at least one". Use the remaining bag for "the first is red".

Q07

Draw the possible waiting times on [0,10]. For the cutoff policy, split arrivals into two equally likely groups.

Q08

A random moment is more likely to land in a long interval. Weight intervals by time, then average the residual wait.

Q09

For a roll r, you win only if both opponents roll below r. For restarts, use eventual win = one-round win / one-round resolution.

Q10

Compare a known first value with the expected value of a fresh roll.

Q11

Spend one roll. On failure, you are in the same state as before.

Q12

Use two states: no useful head so far, and a trailing head. A tail resets progress.

06 / HINTS

One nudge at a time / 2 of 2

Use the smallest hint you need, then return to the question. Do not read ahead for a mock.

Q13

Before both start, handle the head start separately. Once both move, add their closing speeds.

Q14

The grandmother is 10a+b, not a+b. Use the range to test a = 6, 7 or 8.

Q15

A sum of two consecutive integers is odd. For odd-length runs, look at the middle integer.

Q16

Temporarily write every number from 000 to 999 using exactly three positions.

Q17

Distinguish identifying the code after failures from actually entering it successfully. In the follow-up, different insertions may produce the same code.

Q18

Use ordered pairs. After receiving information, retain only pairs consistent with it.

Q19

Dry material is 5% of the starting mass.

Q20

A length-five palindrome is ABCBA. Exactly three distinct symbols means A, B and C are pairwise different.

Q21

The eligible outcomes are GG and OO, but they do not have equal probabilities.

Q22

Uniform residual wait has mean half the interval. Then use the inequality W+6 < 11.

Q23

For a maximum M, P(M <= k) = (k/6)^2. Subtract consecutive squares to get P(M=k).

Q24

Convert 20 minutes to one third of an hour. Work out the head start, then use closing speed.

07 / WORKED SOLUTIONS

Solutions 01-03

Read the reasoning, close the page, then solve a changed-number version.

Q01 The dehydrating strawberry

Answer 50 g. In general, final mass = M(1-p)/(1-q).

The dry material weighs 1 g. It becomes 2% of the final mass, so 0.02m = 1 and m = 50. A one-percentage-point change in water content doubles the dry fraction. The whole mass therefore halves.

Watch for Subtracting 1% from the original mass confuses a percentage with a percentage point. Check that the dry material still weighs 1 g.

Q02 The painted cube

Answer Any of five faces: 25/27. Top face only: 1/3. General case: 1 - (n-2)^2(n-1)/n^3.

There are 8 corners (3 painted faces), 12 edge-middle cubes (2), 6 face-centre cubes (1), and 1 interior cube (0). Paint is invisible only for the interior cube or when the single painted face of a face-centre cube points down. Failure probability is 1/27 + (6/27)(1/6) = 2/27. For the top-face question, 54 of the 162 small faces are painted. For general n, failure count weighted by orientation is (n-2)^3 + (n-2)^2.

Watch for Choosing a cube uniformly is different from choosing only a surface cube. Also clarify whether "see paint" means any visible face or just the top.

Q03 Palindromes: count the free positions

Answer Four letters: 676. Five letters: 17,576. Five with no equal neighbours: 26 x 25 x 25 = 16,250. General: a^ceil(n/2).

ABBA has two freely chosen letters; ABCBA has three. With no equal adjacent letters in ABCBA, choose A in 26 ways, B in 25 ways, and C in 25 ways because C must differ from B, but may equal A. The second half is forced.

Watch for A palindrome is not a permutation of distinct letters. Requiring every letter to be different would make a length greater than one impossible.

07 / WORKED SOLUTIONS

Solutions 04-06

Read the reasoning, close the page, then solve a changed-number version.

Q04 A random three-digit number

Answer Integer model: 2/9. Three-digit-string model: 1/5.

There are 900 three-digit integers. Favourable choices: 5 odd leading digits, 5 even middle digits, and 8 final digits because the first two differ. Thus 5 x 5 x 8 / 900 = 2/9. For independent digit strings there are 1,000 equally likely strings and the same 200 favourable ones.

Watch for Do not use 1,000 as the denominator for a uniformly selected three-digit integer. State the sampling model before calculating.

Q05 Exactly three heads

Answer Exactly three: 5/16. Given at least one tail: 10/31. Biased coin: 10p^3(1-p)^2.

There are C(5,3) = 10 head-position patterns, each of probability 1/32. Conditioning removes only HHHHH, leaving 31 equally likely patterns; all ten three-head patterns remain. For a biased coin, each qualifying sequence has probability p^3(1-p)^2.

Watch for The probability of one particular sequence is not the probability of every ordering with three heads.

Q06 Red and blue balls

Answer Given at least one red: 1/3. Given the first is red: 1/2.

Of C(5,2) = 10 pairs, only one pair is all blue. Of the nine eligible pairs, C(3,2) = 3 are all red, so 3/9 = 1/3. Knowing the first ball is red instead leaves two red balls among four remaining, giving 2/4. The information selects different sets of possible outcomes.

Watch for Treating "at least one is red" as identical to "the first is red" changes the conditioning event. Ball counts here are supplied for practice.

07 / WORKED SOLUTIONS

Solutions 07-09

Read the reasoning, close the page, then solve a changed-number version.

Q07 Wait for the train or walk?

Answer Train and walking both average 10 minutes; train is faster with probability 1/2. The five-minute cutoff policy averages 11.25 minutes.

The wait W is uniform on [0,10], so E[W] = 5 and P(W+5 < 10) = 1/2. Under the cutoff policy, W <= 5 gives average journey 2.5 + 5 = 7.5 minutes. W > 5 gives 5 + 10 = 15 minutes. Each event has probability 1/2, giving 11.25.

Watch for A sensible-sounding fallback can increase expected time. Measure total time from arrival, including time already spent waiting.

Q08 Why irregular buses make you wait

Answer Expected wait: 6.25 minutes. Train/bus journey: 11.25 minutes, so walking has the lower expectation.

In the long run, 5-minute gaps occupy 1/4 of elapsed time and 15-minute gaps occupy 3/4. Conditional on a gap length L, your arrival is uniform within it, so the average residual wait is L/2. Therefore E[W] = (1/4)(2.5) + (3/4)(7.5) = 6.25. Generally, for these stationary renewal assumptions, E[W] = E[L^2]/(2E[L]). Here this is 125/20.

Watch for Choosing a random departure interval and choosing a random point in time are different experiments. The interval model here is an authored extension.

Q09 Three dice and a restart rule

Answer All fair: one-round win 55/216; eventual win 1/3. Fixed 4: one-round win 1/4; eventual win 1/3.

All fair: sum (r-1)^2 over r = 1,...,6 gives 55 winning triples out of 216. Exactly one player wins with probability 3 x 55/216 = 55/72, so eventual win is (55/216)/(55/72) = 1/3. With your 4 fixed, you win in 9 of 36 opponent pairs. Each opponent also uniquely wins in 9 pairs (winning rolls 5 or 6 contribute 4 + 5), leaving 9 tied pairs. Thus (9/36)/(27/36) = 1/3.

Watch for The published recollection mentions only some ties. This version explicitly restarts every tied maximum; other tie rules can change the answer.

07 / WORKED SOLUTIONS

Solutions 10-12

Read the reasoning, close the page, then solve a changed-number version.

Q10 One optional reroll

Answer Reroll 1, 2 or 3; keep 4, 5 or 6. Expected payoff 17/4 = 4.25. Fair option fee: 3/4.

A forced fresh roll averages 3.5. Keep only values above that. Average max(x,3.5) across x = 1,...,6: (3 x 3.5 + 4 + 5 + 6)/6 = 4.25. Without the option the payoff averages 3.5, so the option adds 0.75 in expected value.

Watch for After discarding the first result, you cannot reclaim it. This exercise uses a standard setup also present in the official trading guide; it is not a reported TDOE prompt.

Q11 How long until the first six?

Answer Expected rolls: 6. More than three rolls: 125/216. After ten failures: still 6 additional rolls.

Let E be the expected remaining rolls. E = 1 + (5/6)E, so E = 6. More than three rolls means the first three all fail, with probability (5/6)^3. Independence means past failures do not change future success probability. With success probability p, the expectation is 1/p.

Watch for The die is not "due" a six. Say whether the successful trial is included. This is an original practice variation on recursion.

Q12 Two consecutive heads

Answer Two consecutive heads: 6 flips. Three consecutive heads: 14 flips.

Let E0 be the remaining expectation with no trailing head and E1 with one trailing head. E0 = 1 + E0/2 + E1/2 and E1 = 1 + E0/2. Solving gives E0 = 6. For three heads, add E2: E0 = 1 + (E0+E1)/2; E1 = 1 + (E0+E2)/2; E2 = 1 + E0/2. These give E0 = 14. Disjoint two-flip blocks miss runs crossing block boundaries.

Watch for A successful pair has probability 1/4, but overlapping pairs are not independent trials. Use the current run length as the state.

07 / WORKED SOLUTIONS

Solutions 13-15

Read the reasoning, close the page, then solve a changed-number version.

Q13 Two cyclists approaching each other

Answer Simultaneous: 2 hours, 24 km from A. Delayed: 2.3 hours from the first start, 27.6 km from A.

With simultaneous starts the closing speed is 12 + 18 = 30 km/h, so 60/30 = 2 hours. With the delay, the first cyclist covers 6 km before the second starts. The remaining 54 km closes in 54/30 = 1.8 hours. Total elapsed time is 0.5 + 1.8 = 2.3 hours, and the meeting point is 12 x 2.3 = 27.6 km from A.

Watch for Use a common clock and check both distances add to 60. The speeds and distance are supplied for practice, not quoted from the report.

Q14 Grandmother and two grandchildren

Answer Grandmother: 69. Grandchildren: 6 and 9.

The total is (10a+b)+a+b = 11a+2b = 84. As a is 6, 7 or 8, a=6 gives b=9; a=7 gives a non-integer b; a=8 gives a negative b. Only 69, 6 and 9 satisfy every constraint. The original recollection omits crucial conditions; this is a complete practice version.

Watch for Do not invent missing constraints silently in an interview. Say what extra information is needed before trying to identify a unique answer.

Q15 Consecutive sums

Answer 975 = 487+488 = 324+325+326 = 193+194+195+196+197.

The two-term sum must be odd. A three-term consecutive sum is three times its middle term, so it is divisible by 3. A five-term sum is five times its middle term, so it is divisible by 5. Hence the target must be an odd multiple of 15. The greatest multiple of 15 below 1,000 is 990, which is even; the next is 975. Its displayed positive representations establish sufficiency.

Watch for Necessary conditions alone are not a complete proof. Show the actual runs and verify their sums.

07 / WORKED SOLUTIONS

Solutions 16-18

Read the reasoning, close the page, then solve a changed-number version.

Q16 Counting a digit without listing numbers

Answer Digit 1 appears 300 times. In the ordinary, unpadded list, digit 0 appears 189 times.

Across 000 to 999 each of three positions contains a 1 exactly 100 times, giving 300. Padding adds no ones, so this remains valid for 1 to 999. For zeros, count actual written positions instead: tens-position zero occurs in 100-109, 200-209, ..., 900-909, giving 90. Units-position zero occurs at 10,20,...,990, giving 99. Total: 189.

Watch for Symmetry across padded strings does not automatically survive removal of leading zeros. Define the representation.

Q17 The missing digit

Answer Guaranteed successful entry: 10 trials. Expected trials: 5.5. Unknown deletion position for 12345: 55 distinct candidates.

With a known missing position there are ten equally likely codes, so the correct code is equally likely to be first through tenth in any trial order: mean 5.5. Nine failures identify the digit, but a tenth entry may still be needed. For 12345, ten digits can be inserted in six slots, giving 60 insertion operations. Inserting each existing digit immediately before or after itself gives the same resulting code, causing five duplicate pairs. Thus 60-5 = 55 distinct codes.

Watch for Counting insertion operations is different from counting unique strings. The original report gives only a missing-digit topic.

Q18 Dice: condition before you count

Answer Sum 8: 5/36. A=6 given sum 8: 1/5. Both 6 given at least one 6: 1/11.

Sum 8 corresponds to (2,6),(3,5),(4,4),(5,3),(6,2), so its probability is 5/36. Of those five equally likely pairs, one has A=6. For the last question there are 6 pairs with A=6 and 6 with B=6, with (6,6) counted twice: 11 distinct pairs. Only one is double-six.

Watch for Original independence does not imply conditional independence. Knowing the sum changes the distribution of each die.

07 / WORKED SOLUTIONS

Solutions 19-21

Read the reasoning, close the page, then solve a changed-number version.

Q19 Changing water content

Answer Final mass 100 g; water lost 100 g. At 125 g, water content is 92%.

The batch has 10 g of dry material. At 90% water, dry material is 10% of the whole, giving 10/0.10 = 100 g. Only water was lost, so the mass loss of 100 g is all water. If the final mass is 125 g, water weighs 115 g and its fraction is 115/125 = 0.92.

Watch for Always check dry material in both states; it remains 10 g.

Q20 Constrained palindromes

Answer 125 palindromes; 60 use exactly three symbols. Conditional probability: 12/25.

Choose the free positions A, B and C, giving 5^3 = 125. Requiring three distinct symbols gives 5 x 4 x 3 = 60. All palindromes have one unique free triple, so uniform selection among the 125 palindromes gives 60/125 = 12/25.

Watch for The denominator is the number of palindromes, not all 5^5 strings.

Q21 Information about a sample

Answer Without replacement: 2/3. With replacement: 16/25, which is smaller.

Without replacement, there are C(4,2)=6 green pairs and C(3,2)=3 orange pairs, giving 6/9 = 2/3. With replacement, P(GG)=16/49 and P(OO)=9/49, so the conditional probability is 16/(16+9) = 16/25. Replacement changes the composition after the first draw and reduces the relative advantage of the larger colour group in this example.

Watch for Same-colour pairs are not equally likely merely because there are two colour labels.

07 / WORKED SOLUTIONS

Solutions 22-24

Read the reasoning, close the page, then solve a changed-number version.

Q22 Another commute

Answer Bus expectation: 10 minutes versus 11 walking. Bus faster: 5/8. Just missing a departure gives an 8-minute wait and a 14-minute bus journey.

With W uniform on [0,8], E[W+6]=4+6=10. The bus beats walking when W<5, covering 5 of the 8 minutes in the interval, so probability 5/8. Seeing a departure gives new information: for an exact timetable the next departure is 8 minutes away. Walking is then quicker.

Watch for Update your model when new information arrives instead of repeating an unconditional average.

Q23 The larger die

Answer Two fair dice: 161/36, about 4.4722. One fixed at 4: 9/2 = 4.5.

P(M=k) = [k^2-(k-1)^2]/36 = (2k-1)/36. Summing k(2k-1)/36 for k=1,...,6 gives 161/36. With a fixed 4, the payoff is 4 when the other die is 1,2,3 or 4, then 5 or 6. Thus (4x4+5+6)/6 = 27/6 = 4.5. Both values should lie between a single-die mean of 3.5 and the maximum of 6.

Watch for The maximum of two fair dice is not uniform on 1 to 6.

Q24 A delayed cyclist

Answer Simultaneous: 2 hours, 20 km from the slower rider's start. Delayed: 2.2 hours, 22 km from that start.

The closing speed is 25 km/h, so simultaneous riders meet after 50/25=2 hours. With the delay, let t be hours since the slower rider starts. Distances satisfy 10t + 15(t-1/3)=50, so 25t=55 and t=2.2. The slower rider travels 22 km and the faster travels 28 km. These sum to 50.

Watch for Do not subtract a delay measured in minutes from a time measured in hours.

08 / FINAL REVIEW

What to remember on the day

Your goal is a clear, adaptable solution process. A memorised answer is only a starting point.

BEFORE CALCULATING

State uniformity, independence, replacement and tie rules. Ask what information was actually observed. For a numeric puzzle, check whether enough conditions exist for a unique answer.

WHEN THE ANSWER FEELS COUNTERINTUITIVE

Check the unchanged quantity, the denominator, the units and the sampling method. Long bus gaps attract more random arrivals. Observed conditions change which outcomes remain possible.

WHEN YOU GET STUCK

Try a smaller case, a table, a diagram or an equation for a restart. Share the partial structure you have found. Use a hint to revise the model and then explain the revision.

WHEN YOU FINISH

Check a bound and a simple case. Explain one alternative route. Ask whether the follow-up changes the sample space, payoff or transition rules.

PREPARE TWO SHORT PERSONAL ANSWERS

Why Jane Street? Why TDOE specifically? Connect your own experience to solving practical problems, learning unfamiliar concepts, accuracy and working with other people. Prepare one real example of finding and correcting a mistake. Motivation questions are reported, but their stage varies. [W1]

IF YOU ONLY REVIEW SIX THINGS

1. Dry mass is conserved.
2. Count only free positions.
3. Condition on the information actually received.
4. Average outcomes using their probabilities.
5. A restart is an equation, not a reason to panic.
6. Changed assumptions require a changed model.

Optional later-round bridge: if your recruiter confirms coding is next, practise finding the three or k largest trades, discussing ties, empty inputs and efficiency. Do not let that displace the first-live-round preparation this pack is designed for.

09 / SOURCES AND PROVENANCE

Research register / 1 of 2

Researched 24 October 2026. Click a title to open the source. Candidate reports are evidence of recollections, not verified interview scripts.

O1 Official / TDOE role and interview process

Confirms the take-home, initial problem-solving Zoom, technical Zoom and final round. Primary source for the stage distinction.

O2 Official / Probability & Markets guide

Supporting foundations: counting, expectation, conditioning and recursion. Written for trading/research; it is not a TDOE syllabus. Q10 uses a standard exercise also found here; Q11-Q12 are practice variations.

O3 Official / Trading interview and mock video

Useful demonstration of a collaborative interview with follow-ups. Different role: use for communication practice, not as a promise of TDOE questions.

G0 Glassdoor / original 33 TDOE reports

All seven pages read in the preceding chat: 33 reports, with posting dates from 12 Nov 2024 to 7 Oct 2026. At least two accounts repeat identical wording.

G1 Glassdoor / strawberry and first live maths round

Posted 3 Nov 2025. Lists a strawberry ratio problem and bus waiting time after the Excel task. The 100 g / 99% / 98% version is also described in a 24 Mar 2026 report on page 2 of G0.

G2 Glassdoor / painted cube and full process

Posted 20 Jul 2026; candidate says interview occurred in spring 2024. Describes a painted 3 x 3 x 3 cube and a probability/brain-teaser phone round.

G3 Glassdoor / four-letter palindromes

TDOE Intern question, posted 13 Nov 2025. Alphabet size and follow-ups in this pack are supplied for practice.

G4 Glassdoor / odd-even-distinct digits

TDOE Intern question, posted 19 Feb 2025. Sampling model is made explicit in Q04.

G5 Glassdoor / five coin flips

TDOE Intern question, posted 4 Mar 2025. Conditional and biased-coin follow-ups are authored additions.

G0 coverage: the original listing plus pages 2-7, reached through its pagination. No claim is made that these 33 reports are the complete set under every Jane Street role title.

09 / SOURCES AND PROVENANCE

Research register / 2 of 2

Researched 24 October 2026. Click a title to open the source. Candidate reports are evidence of recollections, not verified interview scripts.

G6 Glassdoor / conditional probability with coloured balls

Posted 26 Aug 2025; interview reportedly Aug 2024. Reports red/blue balls and short probability/ratio questions; counts and precise conditions are reconstructed.

G7 Glassdoor / interactive train waiting problem

Posted 14 Jun 2026, adjacent Trading Desk Operations title. Reports 10-minute train frequency, 5-minute ride, 10-minute walk and evolving follow-ups. Q07-Q08 supply exact arrival models.

G8 Glassdoor / waiting-time paradox report

A report posted 2 Apr 2025 describes an adaptation of the waiting-time paradox. Adjacent title; supporting topic evidence rather than an exact prompt.

G9 Glassdoor / three-player dice and palindromes

Posted 22 Mar 2026; interview Feb 2026. Reports a dice win/restart question and a guaranteed-4 follow-up. Q09 deliberately specifies all maximum-tie handling.

G10 Glassdoor / counting and speed-distance-time

Posted 18 Apr 2026. Explicitly places counting and a motion problem in the first Zoom interview. Q13 numbers are authored.

G11 Glassdoor / age-digit riddle

Posted 14 Mar 2025. The age riddle is incomplete. Q14 adds a range and total to create a uniquely solvable practice version.

G12 Glassdoor / consecutive-integer sums

TDOE Intern question, posted 2 Jun 2025. This pack explicitly requires positive integers and checks all three constructions.

G13 Glassdoor / missing digit and motion topics

Report posted 25 Sep 2026, adjacent title. Mentions missing digits and speed/time. Access-code framing and the 12345 follow-up in Q17 are authored.

W1 Wall Street Oasis / TDOE intern account

New York interview dated Oct 2024. Describes three first-round brainteasers plus motivation questions, followed by coding. Independent platform, still unverified candidate evidence.

W2 Wall Street Oasis / recent TDOE accounts

Visible accounts submitted Jun 2026 corroborate first-round basic probability, strawberry/cube questions and later coding. Used only as corroboration; role labels vary.